

THERMODYNAMICS

16.1 Internal energy

Candidates should be able to:

- 1 understand that internal energy is determined by the state of the system and that it can be expressed as the sum of a random distribution of kinetic and potential energies associated with the molecules of a system
- 2 relate a rise in temperature of an object to an increase in its internal energy

16.2 The first law of thermodynamics

Candidates should be able to:

- 1 recall and use $W = p\Delta V$ for the work done when the volume of a gas changes at constant pressure and understand the difference between the work done by the gas and the work done on the gas
- 2 recall and use the first law of thermodynamics $\Delta U = q + W$ expressed in terms of the increase in internal energy, the heating of the system (energy transferred to the system by heating) and the work done on the system

1. O/N/09/42/Q2

- (a) State what is meant by the *internal energy* of a gas.

.....
.....
..... [2]

- (b) The first law of thermodynamics may be represented by the equation

$$\Delta U = q + w.$$

State what is meant by each of the following symbols.

$+\Delta U$
 $+q$
 $+w$ [3]

- (c) An amount of 0.18 mol of an ideal gas is held in an insulated cylinder fitted with a piston, as shown in Fig. 2.1.

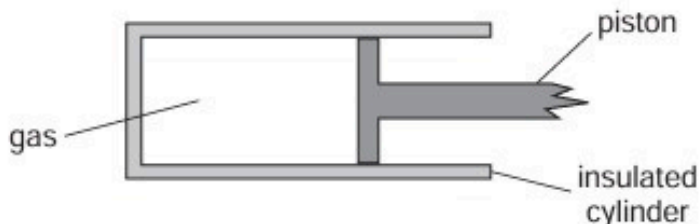


Fig. 2.1

Atmospheric pressure is $1.0 \times 10^5 \text{ Pa}$.

The volume of the gas is suddenly increased from $1.8 \times 10^3 \text{ cm}^3$ to $2.1 \times 10^3 \text{ cm}^3$.

For the expansion of the gas,

- (i) calculate the work done by the gas and hence show that the internal energy changes by 30 J,

2. O/N/10/41/Q2

- (a) (i) State the basic assumption of the kinetic theory of gases that leads to the conclusion that the potential energy between the atoms of an ideal gas is zero.

.....
.....[1]

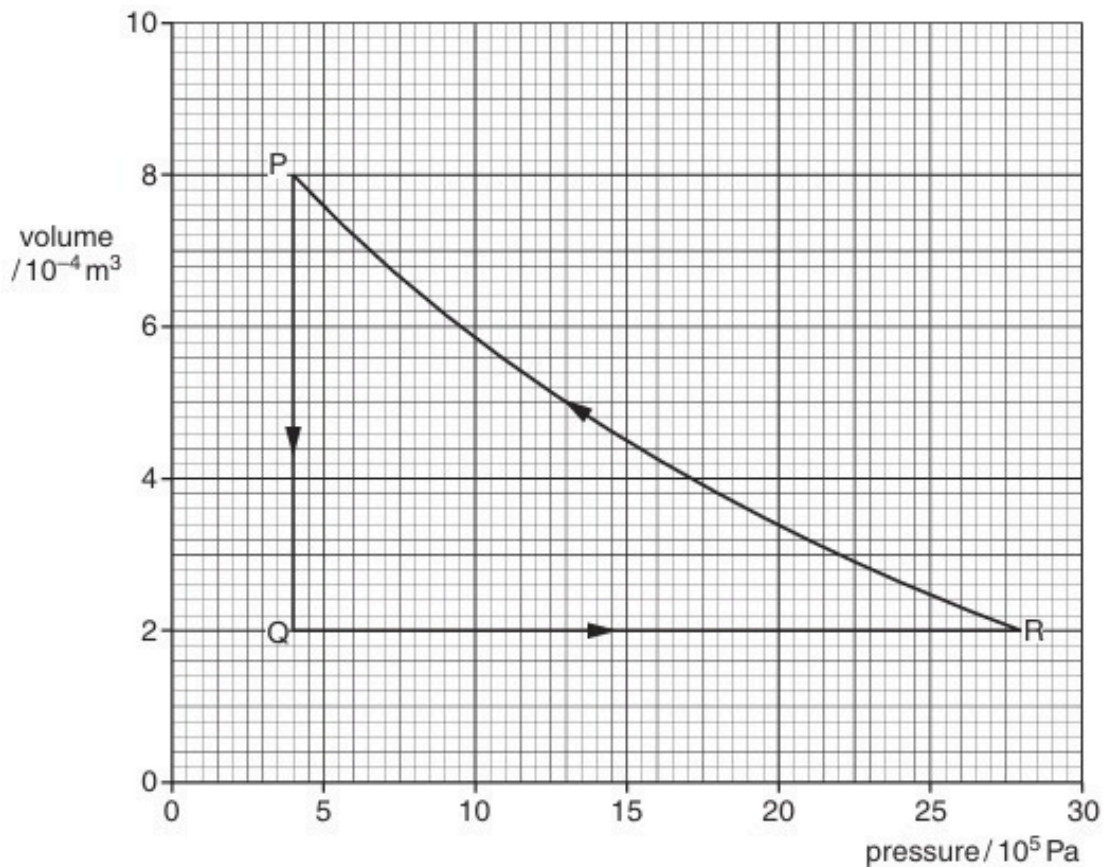
- (ii) State what is meant by the *internal energy* of a substance.

.....
.....
.....[2]

- (iii) Explain why an increase in internal energy of an ideal gas is directly related to a rise in temperature of the gas.

.....
.....
.....[2]

- (b) A fixed mass of an ideal gas undergoes a cycle PQRP of changes as shown in Fig. 2.1.



- (i) State the change in internal energy of the gas during one complete cycle PQRP.

change = J [1]

- (ii) Calculate the work done on the gas during the change from P to Q.

work done = J [2]

- (iii) Some energy changes during the cycle PQRP are shown in Fig. 2.2.

change	work done on gas / J	heating supplied to gas / J	increase in internal energy / J
P → Q		−600	
Q → R	0	+720	
R → P		+480	

Fig. 2.2

Complete Fig. 2.2 to show all of the energy changes.

[3]

3. M/J/11/42/Q4

- (a) The first law of thermodynamics may be expressed in the form

$$\Delta U = q + w.$$

Explain the symbols in this expression.

+ ΔU

+ q

+ w

[3]

- (b) (i) State what is meant by *specific latent heat*.

.....

.....

.....

..... [3]

- (ii) Use the first law of thermodynamics to explain why the specific latent heat of vaporisation is greater than the specific latent heat of fusion for a particular substance.

.....

.....

.....

..... [3]

4. M/J/12/42/Q3

- (a) State what is meant by the *internal energy* of a system.

.....

.....

..... [2]

- (b) State and explain qualitatively the change, if any, in the internal energy of the following systems:

- (i) a lump of ice at 0°C melts to form liquid water at 0°C ,

.....

.....

.....

..... [3]

- (ii) a cylinder containing gas at constant volume is in sunlight so that its temperature rises from 25°C to 35°C .

.....

.....

.....

..... [3]

5. O/N/12/41/Q2

A student suggests that, when an ideal gas is heated from 100°C to 200°C , the internal energy of the gas is doubled.

(a) (i) State what is meant by *internal energy*.

.....
.....
..... [2]

(ii) By reference to one of the assumptions of the kinetic theory of gases and your answer in **(i)**, deduce what is meant by the internal energy of an ideal gas.

.....
.....
.....
..... [3]

(b) State and explain whether the student's suggestion is correct.

.....
.....
..... [2]

6. O/N/13/43/Q2

- (a) (i) State what is meant by the *internal energy* of a system.

.....
.....
..... [2]

- (ii) Explain why, for an ideal gas, the internal energy is equal to the total kinetic energy of the molecules of the gas.

.....
.....
..... [2]

- (b) The mean kinetic energy $\langle E_K \rangle$ of a molecule of an ideal gas is given by the expression

$$\langle E_K \rangle = \frac{3}{2} kT$$

where k is the Boltzmann constant and T is the thermodynamic temperature of the gas.

A cylinder contains 1.0 mol of an ideal gas. The gas is heated so that its temperature changes from 280 K to 460 K.

- (i) Calculate the change in total kinetic energy of the gas molecules.

change in energy = J [2]

- (ii) During the heating, the gas expands, doing $1.5 \times 10^3 \text{ J}$ of work.
State the first law of thermodynamics. Use the law and your answer in (i) to determine the total energy supplied to the gas.

.....

.....

total energy = J [3]

7. M/J/14/41/Q3

The volume of 1.00 kg of water in the liquid state at 100 °C is $1.00 \times 10^{-3} \text{ m}^3$. The volume of 1.00 kg of water vapour at 100 °C and atmospheric pressure $1.01 \times 10^5 \text{ Pa}$ is 1.69 m^3 .

- (a)** Show that the work done against the atmosphere when 1.00 kg of liquid water becomes water vapour is $1.71 \times 10^5 \text{ J}$.

[2]

- (b) (i)** The first law of thermodynamics may be given by the expression

$$\Delta U = +q + w$$

where ΔU is the increase in internal energy of the system.

State what is meant by

- 1.** $+q$,

..... [1]

- 2.** $+w$.

..... [1]

- (ii)** The specific latent heat of vaporisation of water at 100 °C is $2.26 \times 10^6 \text{ J kg}^{-1}$.

A mass of 1.00 kg of liquid water becomes water vapour at 100 °C.

Determine, using your answer in **(a)**, the increase in internal energy of this mass of water during vaporisation.

increase in internal energy = J [2]

8. M/J/15/41/Q2

(a) State what is meant by *internal energy*.

.....

.....

..... [2]

(b) The variation with volume V of the pressure p of an ideal gas as it undergoes a cycle ABCA of changes is shown in Fig. 2.1.

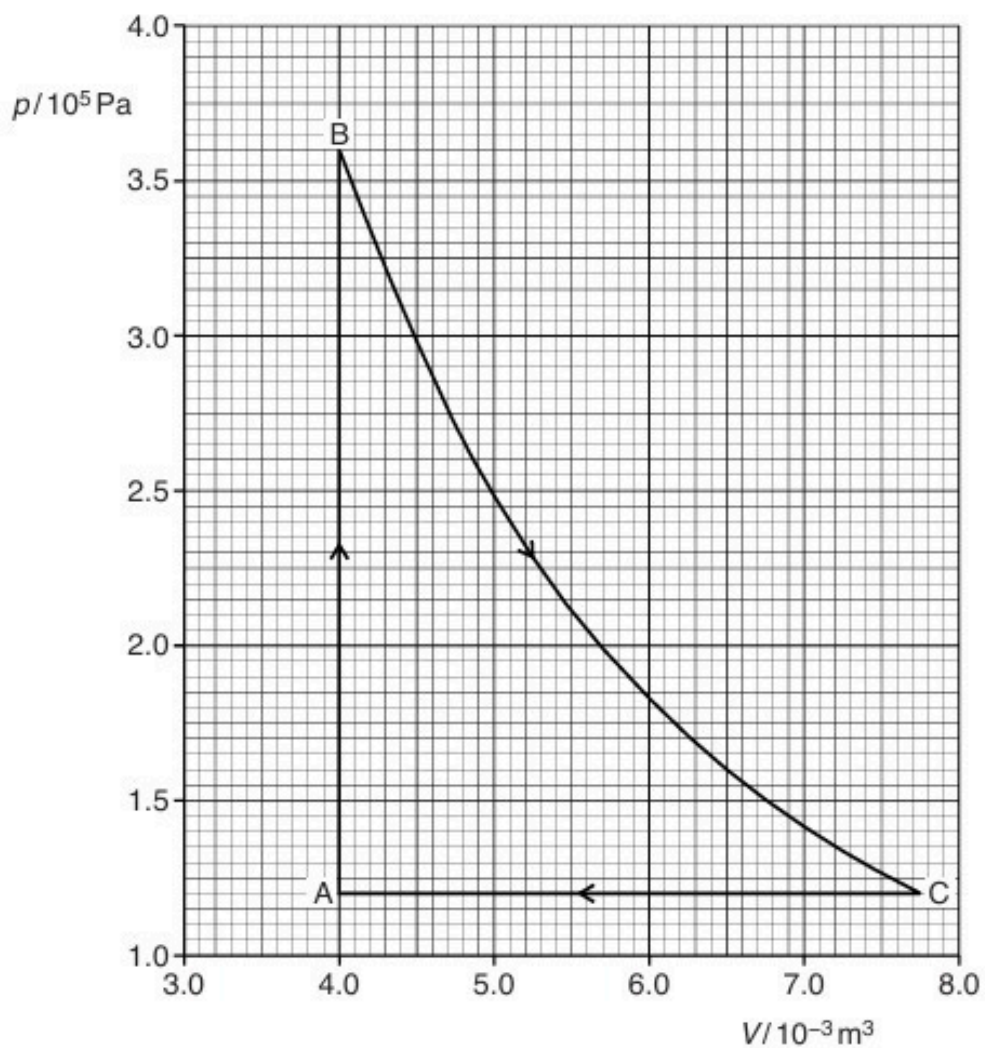


Fig. 2.1

The temperature of the gas at A is 290 K. The temperature at B is 870 K.

Determine

(i) the amount, in mol, of gas,

amount =mol [2]

(ii) the temperature of the gas at C.

temperature = K [2]

(c) Explain why the change from C to A involves external work and a change in internal energy.

.....
.....
..... [2]

9. O/N/15/41/Q3

- (a)** State an expression, in terms of work done and heating, that is used to calculate the increase in internal energy of a system.

.....
.....
.....[2]

- (b)** State and explain, in terms of your expression in **(a)**, the change, if any, in the internal energy

- (i)** of the water in an ice cube when the ice melts, at atmospheric pressure, to form a liquid without any change of temperature,

.....
.....
.....
.....[3]

- (ii)** of the gas in a tyre when the tyre bursts so that the gas suddenly increases in volume. Assume that the gas is ideal.

.....
.....
.....
.....[3]

10. O/N/16/41/Q2

An ideal gas initially has pressure $1.0 \times 10^5 \text{ Pa}$, volume $4.0 \times 10^{-4} \text{ m}^3$ and temperature 300 K , as illustrated in Fig. 2.1.

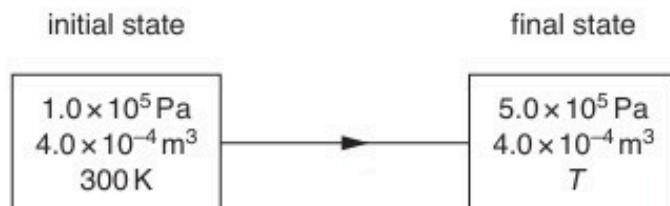


Fig. 2.1

A change in energy of the gas of 240 J results in an increase of pressure to a final value of $5.0 \times 10^5 \text{ Pa}$ at constant volume.

The thermodynamic temperature becomes T .

(a) Calculate

(i) the temperature T ,

$T = \dots\dots\dots \text{ K}$ [2]

(ii) the amount of gas.

amount = $\dots\dots\dots \text{ mol}$ [2]

(b) The increase in internal energy ΔU of a system may be represented by the expression

$$\Delta U = q + w.$$

(i) State what is meant by the symbol

1. $+q$,

.....

2. $+w$.

.....

[2]

(ii) State, for the gas in **(a)**, the value of

1. ΔU ,

$\Delta U =$ J

2. $+q$,

$+q =$ J

3. $+w$.

$+w =$ J

[3]

[Total: 9]

11. O/N/16/42/Q3

- (a) State what is meant by the *internal energy* of a system.

.....

.....

.....[2]

- (b) Explain, by reference to work done and heating, whether the internal energy of the following increases, decreases or remains constant:

- (i) the gas in a toy balloon when the balloon bursts suddenly,

.....

.....

.....

.....[3]

- (ii) ice melting at constant temperature and at atmospheric pressure to form water that is more dense than the ice.

.....

.....

.....

.....[3]

[Total: 8]

12. F/M/17/42/Q2

(a) The first law of thermodynamics can be represented by the expression

$$\Delta U = q + w.$$

State what is meant by the symbols in the expression.

$+\Delta U$

$+q$

$+w$

[2]

(b) A fixed mass of an ideal gas undergoes a cycle ABCA of changes, as shown in Fig.2.1.

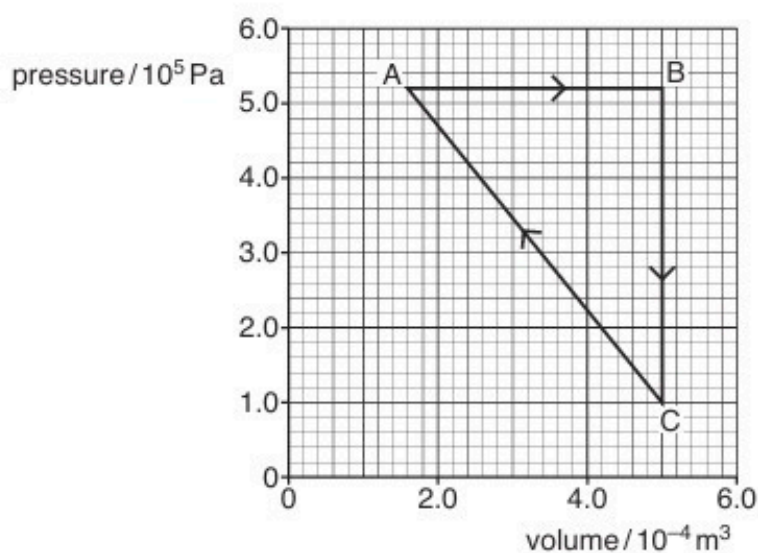


Fig.2.1

(i) During the change from A to B, the energy supplied to the gas by heating is 442 J.

Use the first law of thermodynamics to show that the internal energy of the gas increases by 265 J.

[2]

- (ii) During the change from B to C, the internal energy of the gas decreases by 313 J.

By considering molecular energy, state and explain qualitatively the change, if any, in the temperature of the gas.

.....

.....

.....

.....

.....[3]

- (iii) For the change from C to A, use the data in (b)(i) and (b)(ii) to calculate the change in internal energy.

change in internal energy = J [1]

- (iv) The temperature of the gas at point A is 227 °C. Calculate the number of molecules in the fixed mass of the gas.

number = [2]

[Total: 10]

13. M/J/18/41/Q3

- (a) (i) State what is meant by the *internal energy* of a system.

.....

.....

.....

.....[2]

- (ii) Explain why, for an ideal gas, the change in internal energy is directly proportional to the change in thermodynamic temperature of the gas.

.....

.....

.....

.....

.....[3]

- (b) A cylinder of volume $1.8 \times 10^4 \text{ cm}^3$ contains helium gas at pressure $6.4 \times 10^6 \text{ Pa}$ and temperature 25°C .

Helium gas may be considered to be an ideal gas consisting of single atoms.

Calculate the number of helium atoms in the cylinder.

number =[3]

[Total: 8]

14. O/N/18/41/Q2

(a) State what is meant by the *internal energy* of a system.

.....

.....

.....

.....[2]

(b) An ideal gas undergoes a cycle of changes as shown in Fig. 2.1.

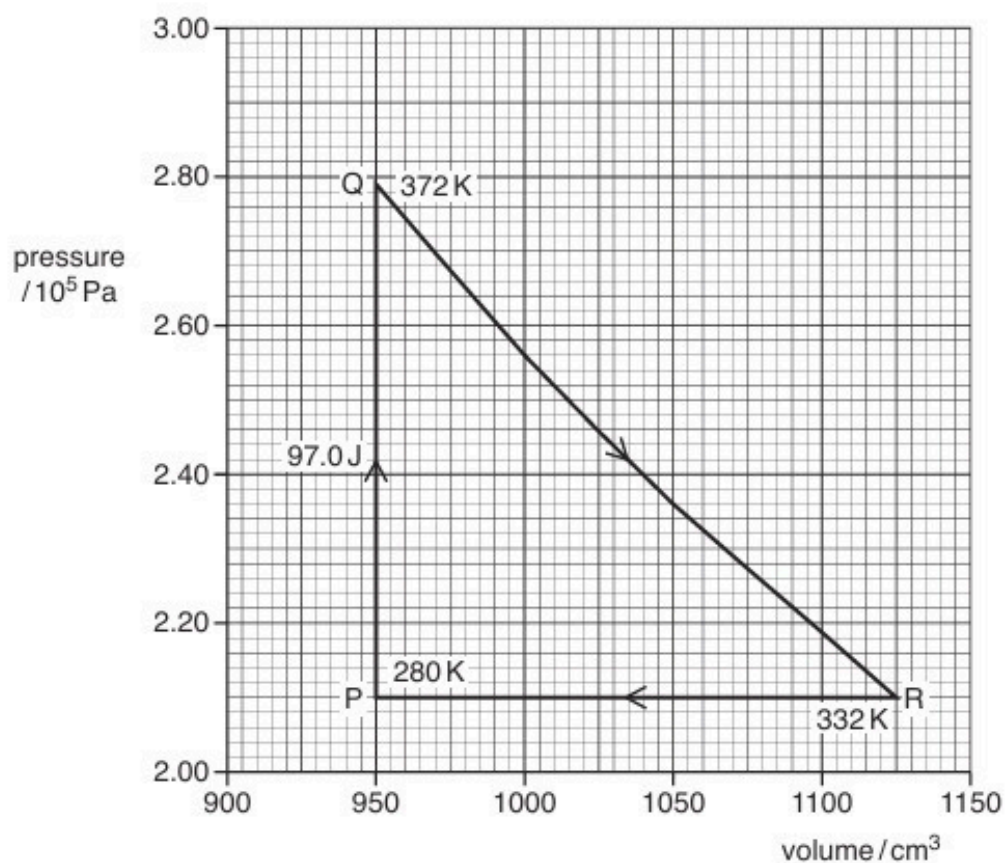


Fig. 2.1

At point P, the gas has volume 950 cm^3 , pressure $2.10 \times 10^5\text{ Pa}$ and temperature 280 K .

The gas is heated at constant volume and 97.0 J of thermal energy is transferred to the gas. Its pressure and temperature change so that the gas is at point Q on Fig. 2.1.

The gas then undergoes the change from point Q to point R and then from point R back to point P, as shown on Fig. 2.1.

Some energy changes that take place during the cycle PQRP are shown in Fig. 2.2.

	change $P \rightarrow Q$	change $Q \rightarrow R$	change $R \rightarrow P$
thermal energy transferred to gas / J	+97.0	0	
work done on gas / J		-42.5	+37.0
increase in internal energy of gas / J			

Fig. 2.2

- (i) State the total change in internal energy of the gas during the complete cycle PQRP. Explain your answer.

.....

[2]

- (ii) On Fig. 2.2, complete the energy changes for the gas during

1. the change $P \rightarrow Q$,
2. the change $Q \rightarrow R$,
3. the change $R \rightarrow P$.

[5]

[Total: 9]

15. M/J/19/42/Q2

(a) The first law of thermodynamics may be expressed in the form

$$\Delta U = q + w.$$

(i) State, for a system, what is meant by:

1. $+q$

.....
.....

2. $+w$.

.....
.....

[2]

(ii) State what is represented by a negative value of ΔU .

.....
.....

[1]

(b) An ideal gas, sealed in a container, undergoes the cycle of changes shown in Fig. 2.1.

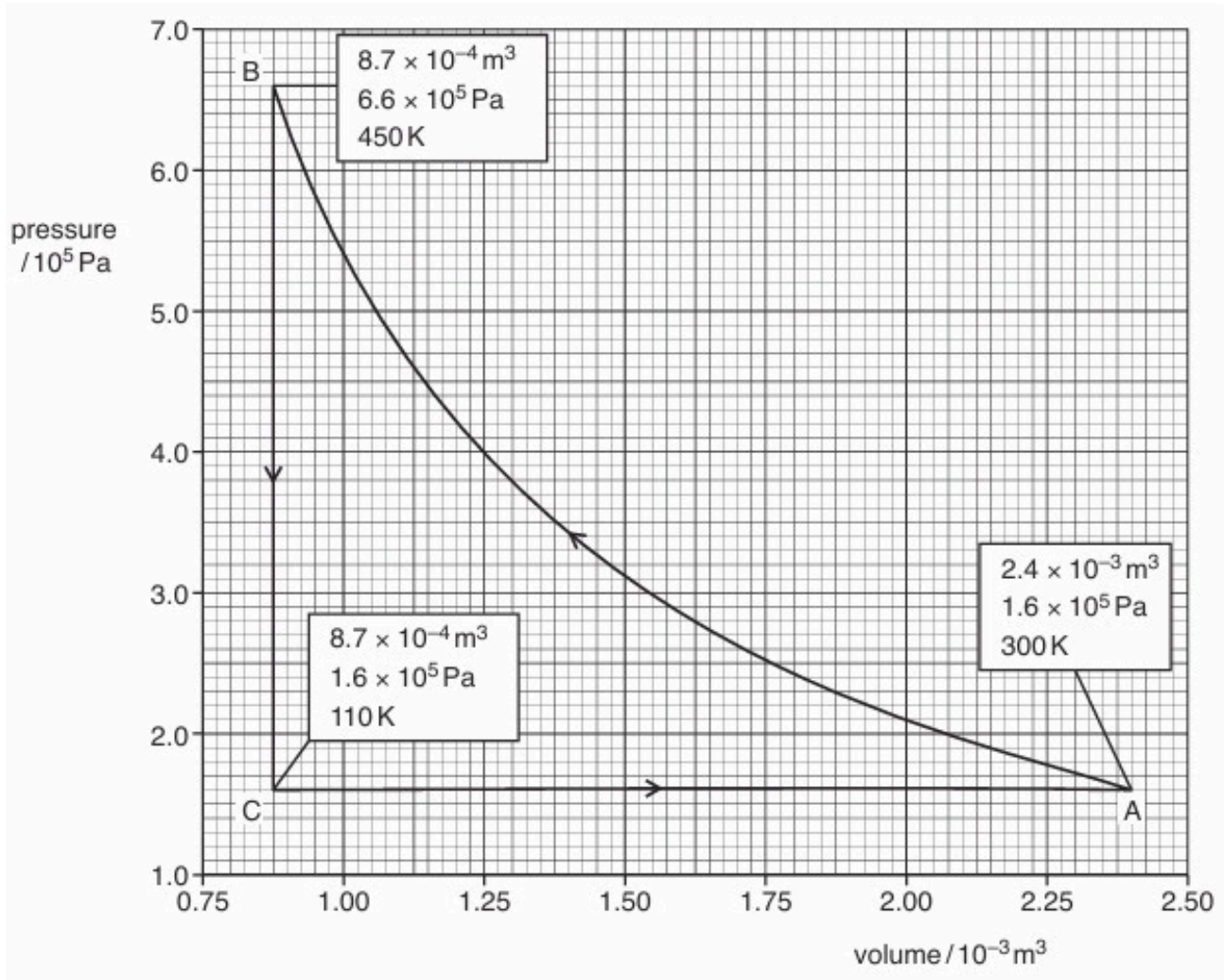


Fig. 2.1

At point A, the gas has volume $2.4 \times 10^{-3} \text{ m}^3$, pressure $1.6 \times 10^5 \text{ Pa}$ and temperature 300 K.

The gas is compressed suddenly so that no thermal energy enters or leaves the gas during the compression. The amount of work done is 480 J so that, at point B, the gas has volume $8.7 \times 10^{-4} \text{ m}^3$, pressure $6.6 \times 10^5 \text{ Pa}$ and temperature 450 K.

The gas is now cooled at constant volume so that, between points B and C, 1100 J of thermal energy is transferred. At point C, the gas has pressure $1.6 \times 10^5 \text{ Pa}$ and temperature 110 K.

Finally, the gas is returned to point A.

- (i) State and explain the total change in internal energy of the gas for one complete cycle ABCA.

.....

[2]

- (ii) Calculate the external work done on the gas during the expansion from point C to point A.

work done = J [2]

- (iii) Complete Fig. 2.2 for the changes from:

1. point A to point B
2. point B to point C
3. point C to point A.

change	$+q/J$	$+w/J$	$\Delta U/J$
A $\rightarrow$ B			
B $\rightarrow$ C			
C $\rightarrow$ A			

Fig. 2.2 [4]

[Total: 11]

16. M/J/20/41/Q2

- (a) State what is meant by the *internal energy* of a system.

.....

.....

.....

..... [2]

- (b) By reference to intermolecular forces, explain why the change in internal energy of an ideal gas is equal to the change in total kinetic energy of its molecules.

.....

.....

..... [2]

- (c) State and explain the change, if any, in the internal energy of a solid metal ball as it falls under gravity in a vacuum.

.....

.....

.....

..... [3]

[Total: 7]

17. M/J/20/42/Q3

By reference to the first law of thermodynamics, state and explain the change, if any, in the internal energy of:

- (a) a lump of solid lead as it melts at constant temperature

.....

.....

.....

.....

..... [3]

- (b) some gas in a toy balloon when the balloon bursts and no thermal energy enters or leaves the gas.

.....

.....

.....

.....

..... [3]

[Total: 6]

18. O/N/20/41/Q2

- (a) The first law of thermodynamics may be expressed as

$$\Delta U = (+q) + (+w)$$

where ΔU is the increase in internal energy of the system.

State the meaning of:

$+q$

.....

$+w$

.....

[2]

- (b) The variation with pressure p of the volume V of a fixed mass of an ideal gas is shown in Fig. 2.1.

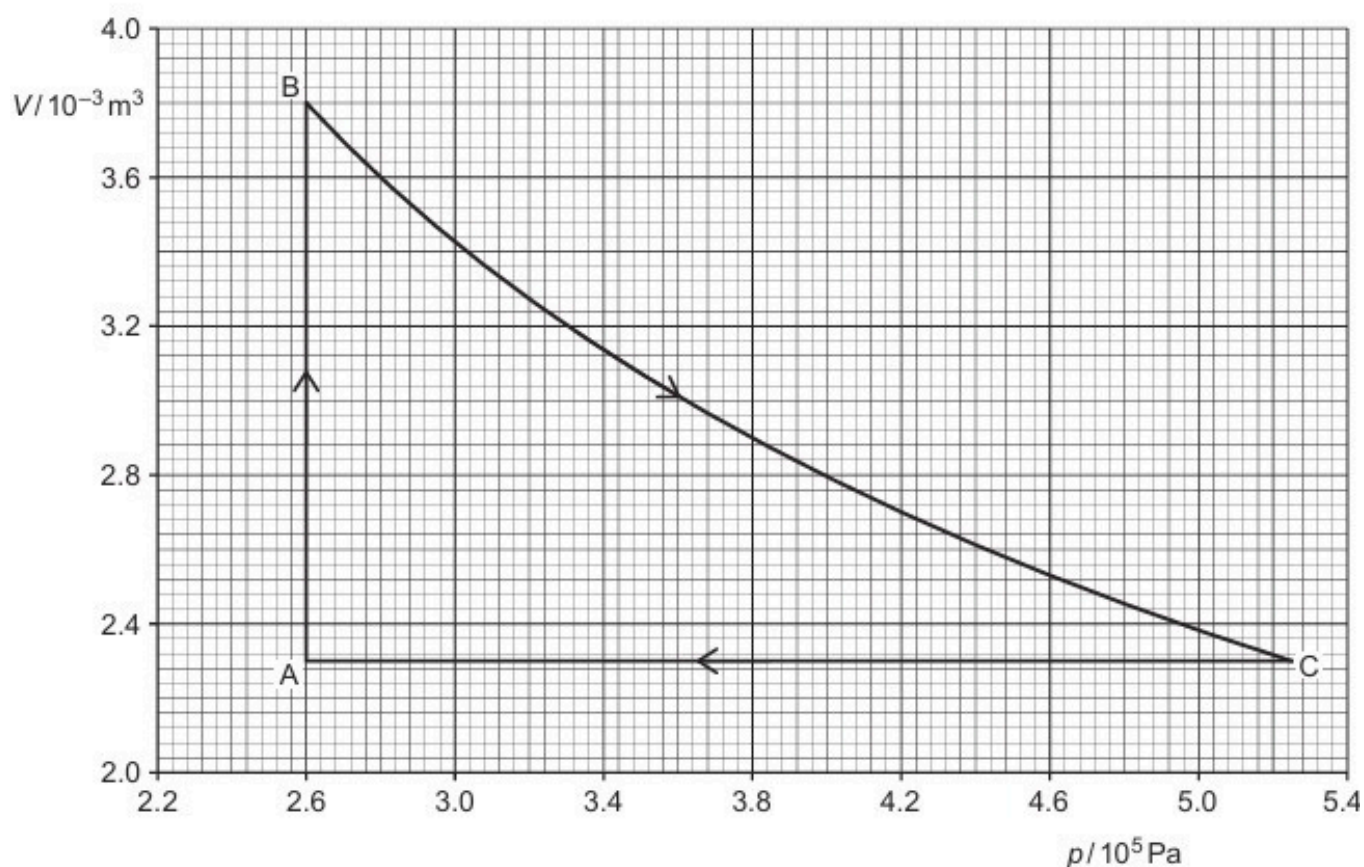


Fig. 2.1

The gas undergoes a cycle of changes A to B to C to A.

During the change A to B, the volume of the gas increases from $2.3 \times 10^{-3} \text{ m}^3$ to $3.8 \times 10^{-3} \text{ m}^3$.

- (i) Show that the magnitude of the work done during the change A to B is 390 J.

[1]

- (ii) State and explain the total change, if any, in the internal energy of the gas during one complete cycle.

.....

.....

..... [2]

- (c) During the change A to B, 1370 J of thermal energy is transferred to the gas.

During the change B to C, no thermal energy enters or leaves the gas. The work done on the gas during this change is 550 J.

Use these data and the information in (b) to complete Table 2.1.

Table 2.1

change	q/J	w/J	$\Delta U/\text{J}$
A to B			
B to C			
C to A			

[4]

[Total: 9]

19. O/N/21/41/Q3

- (a)** Define *specific heat capacity*.

.....
.....
..... [2]

- (b)** A sealed container of fixed volume V contains N molecules, each of mass m , of an ideal gas at pressure p .

- (i)** State an expression, in terms of V , N , p and the Boltzmann constant k , for the thermodynamic temperature T of the gas.

..... [1]

- (ii)** Show that the mean translational kinetic energy E_K of a molecule of the gas is given by

$$E_K = \frac{3}{2} kT.$$

[2]

- (iii)** Explain why the internal energy of the gas is equal to the total kinetic energy of the molecules.

.....
.....
..... [2]

- (c)** The gas in **(b)** is supplied with thermal energy Q .

- (i)** Explain, with reference to the first law of thermodynamics, why the increase in internal energy of the gas is Q .

.....
.....
..... [2]

- (ii) Use the expression in (b)(ii) and the information in (c)(i) to show that the specific heat capacity c of the gas is given by

$$c = \frac{3k}{2m}.$$

[2]

- (d) The container in (b) is now replaced with one that does not have a fixed volume. Instead, the gas is able to expand, so that the pressure of the gas remains constant as thermal energy is supplied.

Suggest, with a reason, how the specific heat capacity of the gas would now compare with the value in (c)(ii).

.....

.....

.....

..... [2]

[Total: 13]

20. F/M/22/42/Q2

A fixed mass of an ideal gas has a volume V and a pressure p .

The gas undergoes a cycle of changes, X to Y to Z to X, as shown in Fig. 2.1.

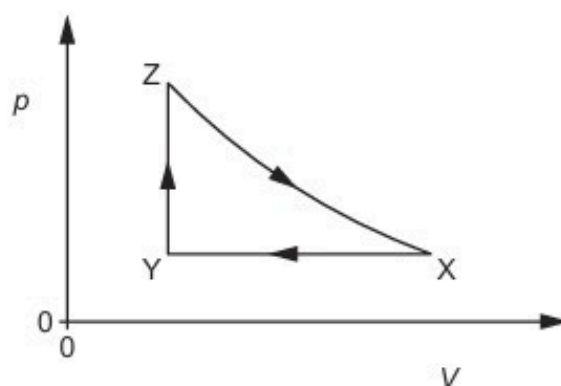


Fig. 2.1

Table 2.1 shows data for p , V and temperature T for the gas at points X, Y and Z.

Table 2.1

	$p / 10^5 \text{ Pa}$	$V / 10^{-3} \text{ m}^3$	T / K
X	1.5	4.2	540
Y			230
Z	5.1		782

(a) State the change in internal energy ΔU for one complete cycle, XYZX.

$\Delta U = \dots\dots\dots \text{ J [1]}$

(b) Calculate the amount n of gas.

$n = \dots\dots\dots \text{ mol [2]}$

- (c) Complete Table 2.1.
Use the space below for any working.

[2]

- (d) (i) The first law of thermodynamics for a system may be represented by the equation

$$\Delta U = q + W.$$

State, with reference to the system, what is meant by:

ΔU :

q :

W :

[3]

- (ii) Explain how the first law of thermodynamics applies to the change Z to X.

.....

.....

.....

..... [2]

[Total: 10]

21. M/J/22/41/Q3

A fixed mass of an ideal gas is initially at a temperature of 17°C .
The gas has a volume of 0.24 m^3 and a pressure of $1.2 \times 10^5\text{ Pa}$.

- (a) (i) State what is meant by an ideal gas.

.....
.....
..... [2]

- (ii) Calculate the amount n of gas.

$n = \dots\dots\dots \text{ mol}$ [2]

- (b) The gas undergoes three successive changes, as shown in Fig. 3.1.

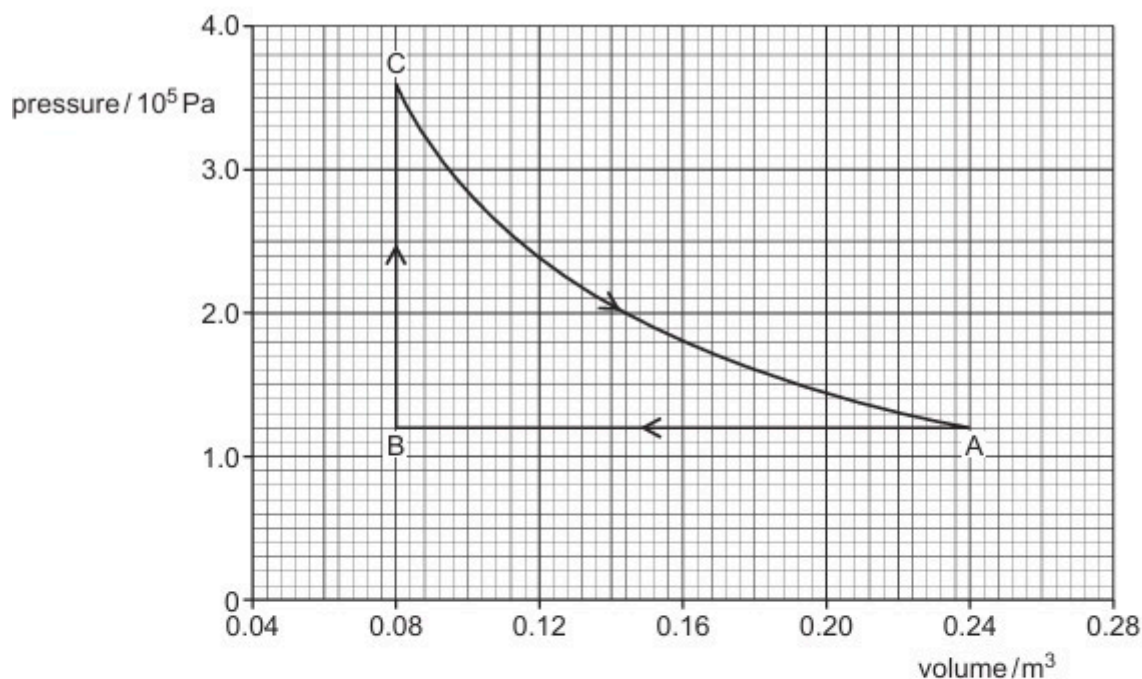


Fig. 3.1

The initial state is represented by point A. The gas is cooled at constant pressure to point B by the removal of 48.0 kJ of thermal energy.

The gas is then heated at constant volume to point C.

Finally, the gas expands at constant temperature back to its original pressure and volume at point A. During this expansion, the gas does 31.6 kJ of work.

- (i) Show that the magnitude of the work done during the change AB is 19.2 kJ.

[2]

- (ii) Complete Table 3.1 to show the work done on the gas, the thermal energy supplied to the gas and the increase in internal energy of the gas, for each of the changes AB, BC and CA.

Table 3.1

change	work done on gas / kJ	thermal energy supplied to gas / kJ	increase in internal energy of gas / kJ
AB		-48.0	
BC			
CA	-31.6		

[5]

[Total: 11]

22. O/N/22/42/Q2

- (a) Define specific heat capacity.

.....

.....

..... [2]

- (b) A fixed mass of water in a beaker is at atmospheric pressure.

- (i) The initial temperature of the water is 0°C .

The water is supplied with thermal energy E , so that its temperature increases to 8°C . There is no net change in the volume of the water.

Use the first law of thermodynamics to complete Table 2.1 for this process.

Table 2.1

work done on water	thermal energy supplied to water	increase in internal energy of water
	$+ E$	

[2]

- (ii) The water is now heated so that its temperature increases by a further 8°C to a final temperature of 16°C . This process causes the volume of the water to increase so that work W is done.

Assume that the change in internal energy is the same as in (b)(i).

Use the first law of thermodynamics to complete Table 2.2 for this process.

Table 2.2

work done on water	thermal energy supplied to water	increase in internal energy of water

[2]

- (c) Use the information in (b) to suggest, with a reason, how the average specific heat capacity of water between 8°C and 16°C compares with its average value between 0°C and 8°C .

.....

..... [1]

[Total: 7]

23. F/M/23/42/Q2

- (a) State what is meant by an ideal gas.

.....

.....

..... [2]

- (b) A fixed amount of helium gas is sealed in a container. The helium gas has a pressure of $1.10 \times 10^5 \text{ Pa}$, and a volume of 540 cm^3 at a temperature of 27°C .

The volume of the container is rapidly decreased to 30.0 cm^3 . The pressure of the helium gas increases to $6.70 \times 10^6 \text{ Pa}$ and its temperature increases to 742°C , as illustrated in Fig. 2.1.

initial state	final state
$1.10 \times 10^5 \text{ Pa}$	$6.70 \times 10^6 \text{ Pa}$
540 cm^3	30.0 cm^3
27°C	742°C

Fig. 2.1

No thermal energy enters or leaves the helium gas during this process.

- (i) Show that the helium gas behaves as an ideal gas.

[2]

- (ii) The first law of thermodynamics may be expressed as

$$\Delta U = q + W.$$

Use the first law of thermodynamics to explain why the temperature of the helium gas increases.

.....

.....

.....

.....

..... [2]

- (iii) The average translational kinetic energy E_K of a molecule of an ideal gas is given by

$$E_K = \frac{3}{2} kT$$

where k is the Boltzmann constant and T is the thermodynamic temperature.

Calculate the change in the total kinetic energy of the molecules of the helium gas.

change in kinetic energy = J [3]

- (c) The mass of nitrogen gas in another container is 24.0 g at a temperature of 27 °C. The gas is cooled to its boiling point of –196 °C. Assume all the gas condenses to a liquid.

For this change the specific heat capacity of nitrogen gas is 1.04 kJ kg^{–1} K^{–1}.

The specific latent heat of vaporisation of nitrogen is 199 kJ kg^{–1}.

Determine the thermal energy, in kJ, removed from the nitrogen gas.

energy = kJ [3]

[Total: 12]

24. M/J/23/41/Q4

(a) State **two** of the basic assumptions of the kinetic theory of gases.

- 1
-
- 2
-

[2]

(b) An ideal gas has amount of substance n .
 The gas is initially in state X, with pressure $2p$ and volume V .
 The gas is cooled at constant volume to state Y, with pressure p .
 The gas is then heated at constant pressure to state Z, with volume $2V$.
 Finally, the gas returns at constant temperature to state X.

- (i) Determine an expression for the temperature T of the gas in state X, in terms of n , p and V .
 Identify any other symbols that you use.

[2]

- (ii) On Fig. 4.1, sketch the variation with volume of pressure for the gas as the gas undergoes the three changes. The state X is labelled. Label states Y and Z.

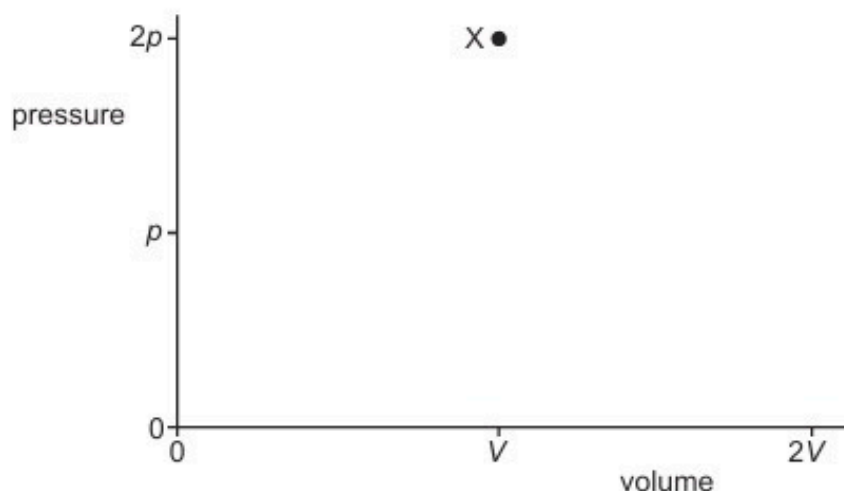


Fig. 4.1

[3]

- (iii) During the change of state from Y to Z, the increase in internal energy of the gas is U .
During the change of state from Z to X, the work done on the gas is W .

Complete Table 4.1 to indicate, for each of the three changes of state, the increase in internal energy of the gas, the thermal energy transferred to the gas and the work done on the gas, in terms of p , V , U and W .

Table 4.1

change	increase in internal energy of gas	thermal energy transferred to gas	work done on gas
X to Y			
Y to Z	$+U$		
Z to X			$+W$

[5]

[Total: 12]

25. M/J/23/42/Q3

(a) State the first law of thermodynamics. Identify the meaning of any symbols that you use.

.....

.....

..... [2]

(b) The state of an ideal gas is continuously changed according to the cycle ABCDA shown in Fig. 3.1.

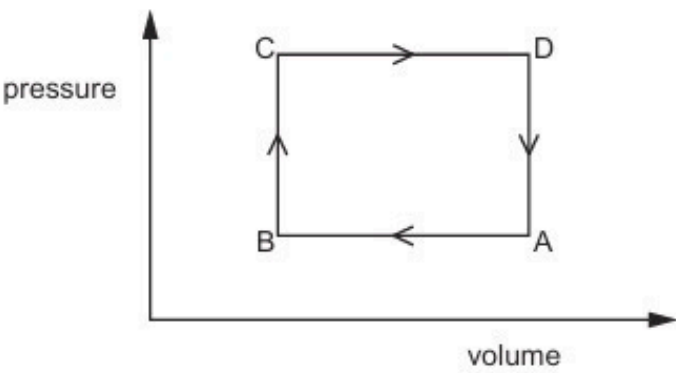


Fig. 3.1

(i) Complete Table 3.1 for the changes A to B and B to C by placing two ticks (✓) in each row.

Table 3.1

change	change in internal energy			work done on gas		
	decrease	no change	increase	negative	zero	positive
A to B						
B to C						

[4]

- (ii) Use the first law of thermodynamics to describe and explain the energy transfers associated with one complete cycle ABCDA.

.....

.....

.....

.....

..... [3]

[Total: 9]

26. O/N/23/41/Q2

- (a) Define specific heat capacity.

.....

.....

..... [2]

- (b) An ideal gas of mass 0.35 kg is heated at a constant pressure of $2.0 \times 10^5 \text{ Pa}$ so that its internal energy increases by 7600 J. During this process, the volume of the gas increases from 0.038 m^3 to 0.063 m^3 and the temperature increases by 56°C .

- (i) Show that the magnitude of the work done on the gas is 5000 J.

[1]

- (ii) Explain whether the work done on the gas is positive or negative.

.....

.....

..... [2]

- (iii) Determine the magnitude of the thermal energy q transferred to the gas.

$$q = \dots\dots\dots \text{ J [2]}$$

- (iv) Calculate the specific heat capacity of the gas for this process. Give a unit with your answer.

$$\text{specific heat capacity} = \dots\dots\dots \text{ unit } \dots\dots\dots [2]$$

- (c) The gas in (b) is now heated at constant volume rather than at constant pressure. The increase in internal energy of the gas is the same as in (b).

Use the first law of thermodynamics to explain whether the specific heat capacity of the gas for this process is less than, the same as, or greater than the answer in (b)(iv).

.....

.....

.....

.....

..... [3]

[Total: 12]

27. O/N/23/42/Q3

(a) State what is meant by the internal energy of a system.

.....

.....

..... [2]

(b) Use the first law of thermodynamics to explain what happens to the internal energy:

(i) of a spring when it is stretched at constant temperature within its elastic limit

.....

.....

.....

.....

..... [3]

(ii) of a sample of water when it evaporates from a rain puddle on a hot day.

.....

.....

.....

.....

..... [3]

[Total: 8]